

1. Administrative & Study Identification

Full Study Title: A Study to Compare the Efficacy and Safety of Entrectinib and Crizotinib in Participants With Advanced or Metastatic ROS1 Non-small Cell Lung Cancer (NSCLC) With and Without Central Nervous System (CNS) Metastases **Short Title/ Acronym:** Entrectinib vs Crizotinib in ROS1+ NSCLC **Protocol Number:** M041552 **NCT Number:** NCT04603807 **Sponsor Name:** Roche (Hoffmann-La Roche) **Study Status:** Recruiting **Investigational Product:** Entrectinib (Rozlytrek) and Crizotinib (Xalkori) **Phase of Development:** Phase III **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To directly compare the efficacy and safety of entrectinib versus crizotinib by evaluating the progression-free survival (PFS) in participants with advanced or metastatic ROS1-positive NSCLC who have central nervous system (CNS) metastases at baseline. **Secondary Objectives:** To evaluate progression-free survival in the Central Nervous System (CNS-PFS), overall response rate (ORR), and duration of response (DOR).

2.2 Background & Rationale To address the unmet need for ROS1-positive NSCLC patients who are at high risk of developing brain metastases. While crizotinib is a standard first-line tyrosine kinase inhibitor (TKI) for ROS1+ NSCLC, it has limited CNS penetration, and many patients experience relapse due to brain metastases or resistance mutations. Entrectinib, a potent ROS1 inhibitor, has demonstrated higher CNS activity. This head-to-head trial aims to test the hypothesis that entrectinib provides superior intracranial and systemic efficacy compared to crizotinib.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Crizotinib) **Blinding:** Open-label **Randomization:** 1:1, Multicenter

3.2 Planned Enrollment & Duration Target \$N\$: 220 patients **Number of Sites:** ~80 sites globally (across at least 10 countries) **Estimated Study Start:** October 2021 **Estimated Primary Completion:** Up to 7 years

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologically or cytologically-confirmed diagnosis of advanced or recurrent (Stage IIIB/C not amenable for radical treatment) or metastatic (Stage IV) NSCLC that harbors a documented ROS1 gene rearrangement. 2. No prior treatment with a ROS1 tyrosine kinase inhibitor, chemotherapy or other systemic therapy for advanced or recurrent (Stage IIIB/C not amenable for radical treatment) or metastatic (Stage IV) NSCLC. 3. Prior radiotherapy is allowed if more than 14 days have elapsed between the end of treatment and randomization. 4. Measurable systemic disease according to RECIST v1.1. 5. Participants with measurable and non-measurable CNS lesions per RECIST v1.1, including leptomeningeal carcinomatosis. 6. Life expectancy of at least 12 weeks. 7. Eastern Cooperative Oncology Group (ECOG) performance status of 0, 1 or 2. 8. Adequate hematologic, renal, liver functions. 9. Participants must have recovered from effects of any major surgery or significant traumatic injury at least 28 days before the first dose of study treatment. 10. Ability to swallow entrectinib and crizotinib intact without chewing, crushing, or opening the capsules. 11. For women of childbearing potential: agreement to remain abstinent (refrain from heterosexual intercourse) or use contraceptive methods with a failure rate of <1% per year during the treatment period and for up to 5 weeks after the last dose of entrectinib or for at least 90 days after the last dose of crizotinib. 12. For men: agreement to remain abstinent (refrain from heterosexual intercourse) or use contraceptive measures, and agreement to refrain from donating sperm.

4.2 Exclusion Criteria 1. Prior treatment with a ROS1 tyrosine kinase inhibitor, chemotherapy or other systemic therapy for advanced or recurrent (Stage IIIB/C not amenable for radical treatment) or metastatic (Stage IV) NSCLC. 2. NCI-CTCAE v5.0 Grade 3 or higher toxicities due to any prior therapy (excluding alopecia, fatigue, nausea and lack of appetite), which have not shown improvement and are strictly considered to interfere with current study drug. 3. History of recent (within the past 3 months) symptomatic congestive heart failure or ejection fraction $\leq 50\%$ observed during screening for the study. 4. History of prolonged corrected QTc interval. 5. Peripheral sensory neuropathy $\geq$ Grade 2. 6. Known interstitial lung disease, interstitial fibrosis, or history of tyrosine kinase inhibitor-induced pneumonitis. 7. Previous malignancy within the past 3 years. 8. Incomplete recovery from any surgery prior to the start of study treatment. 9. Active GI disease (e.g., Crohn's disease, ulcerative colitis or short gut syndrome) or other malabsorption syndrome that would reasonably impact drug absorption. 10. History of prior therapy-induced pneumonitis. 11. Any condition (in the past 3 months) e.g., myocardial infarction, unstable angina, coronary/peripheral artery bypass graft, cerebrovascular accident or transient ischemic attack, stroke, symptomatic bradycardia, or uncontrolled arrhythmias requiring medication.

12. Known active infections (bacterial, fungal or viral, including human immunodeficiency virus positive). 13. History of hypersensitivity to any of the additives in the entrectinib and/or crizotinib drug formulations. 14. Pregnant or lactating women. 15. Known human immunodeficiency virus (HIV) positivity or acquired immunodeficiency syndrome (AIDS)-related illness. 16. Any clinically significant concomitant disease or condition that could interfere with, or for which the treatment might interfere with, the conduct of the study or the absorption of oral medications.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: - *Experimental Arm:* Entrectinib 600 mg once daily (administered as three 200 mg capsules). - *Comparator Arm:* Crizotinib 250 mg twice daily. **Route of Administration:** Oral (with or without food, swallowed intact). **Duration of Treatment:** Treatments will continue until progressive disease, unacceptable toxicity, death, or withdrawal from the study, whichever occurs first.

5.2 Concomitant & Prohibited Therapy Permitted: Prior radiotherapy if more than 14 days have elapsed between the end of treatment and randomization. **Prohibited:** Concurrent administration of other investigational or systemic anti-cancer therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, baseline tumor assessments (brain MRI and chest/abdomen/pelvis CT/MRI), Labs, cardiac function (ejection fraction)
Baseline	Day 1	Randomization, First Dose
Interim	Weeks 4, 8, & every 8 weeks	Tumor assessments via imaging, safety labs, efficacy assessment, Quality of Life questionnaires
End of Study	4 weeks post-treatment	Final Vital Signs, Exit Interview, IP Accountability, Follow-up assessments

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression-free survival (PFS) in participants with central nervous system (CNS) metastases at baseline. PFS is defined as the time from randomization to the first documented disease progression (extracranial or intracranial) or death from any cause whichever occurs first. **Measurement Method:** Determined by a blinded independent review committee (BIRC) using Response Evaluation Criteria in Solid Tumors, Version 1.1 (RECIST v1.1).

7.2 Secondary & Exploratory Endpoints 1. Progression-free survival in the Central Nervous System (CNS-PFS). 2. Overall response rate (ORR). 3. Duration of response (DOR).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of adverse events collected at every visit (blood samples and clinical exams). **Serious Adverse Events (SAEs):** Routine reporting guidelines; SAEs defined as events resulting in death, life-threatening scenarios, or requiring hospitalization. **Data Monitoring Committee (DMC):** External Independent Data Monitoring Committee appointed by the sponsor to oversee trial safety data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for general efficacy outcomes; CNS population (patients with baseline CNS metastases) for the primary endpoint. **Sample Size Justification:** The study is designed to enroll ~220 patients to detect a meaningful difference in the primary endpoint. **Handling of Missing Data:** Standard survival analysis methodologies for handling time-to-event data (e.g., Kaplan-Meier estimation and appropriate censoring for drop-outs).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular site visits (onsite and remote) to monitor investigator compliance, eCRF data entry, and RECIST v1.1 adherence. **Audit Trail:** Robust eCRF locking, tracking of protocol deviations, and image data verification by BIRC.

11. Study Identifiers & Governance

Primary ID: M041552 **Registry Number:** NCT04603807 **Sponsor/Lead Organization:** Roche **Study Title:** Entrectinib vs Crizotinib in ROS1+ NSCLC **Official Regulatory Title:** Randomized, Open Label, Multicenter, Phase III Study of Entrectinib Versus Crizotinib in Patients With Locally-Advanced or Metastatic Non-Small Cell Lung Cancer Harboring ROS1 Gene Rearrangements With and Without Central Nervous System Metastases **Disease Areas:** Carcinoma, Non-Small-Cell Lung

12. Clinical Framework (Hyperkalemia Specific)

Note: This section from the template is N/A for this oncology trial. **Primary Condition:** N/A (Focus is ROS1+ NSCLC) **Diagnosis Threshold:** N/A **Medical Methodology:** N/A **Optimization Target:** N/A

13. Study Procedures & Methodology

Note: This section from the template is specific to a Quality Improvement trial and is N/A here. **Management Pathway:** Standard oncological clinical pathway for targeted therapies. **Education Protocol (HCP):** N/A **Education Protocol (Patient):** N/A **Quality Audit Mechanism:** Independent imaging review via BIRC.

14. Observation & Follow-Up Schedule

Note: Adapted for oncology standard. **Total Duration:** Up to 7 Years **Onsite Visit Cadence:** Baseline, Weeks 4, 8, and every 8 weeks thereafter **Remote Monitoring Frequency:** Intermittent per site-level protocol **Checklist Review Interval:** Scans and assessments every 8 weeks (Q8W)

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				